Problem H4

Møller Scattering: $e^-e^- \to e^-e^-$

Griffiths Problem 7.37 — Spin-Averaged Amplitude & Cross Section

Step-by-step solution with background review

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§1 The Problem (Reworded)

H4 — Problem Statement

Process: Elastic electron–electron (Møller) scattering, $e^-(p_1) + e^-(p_2) \to e^-(p_3) + e^-(p_4)$, in the **massless limit** ($m_e = 0$).

Because the two outgoing electrons are identical, there are **two Feynman diagrams** whose amplitudes are called $\mathcal{M}_1$ (direct/t-channel) and $\mathcal{M}_2$ (exchange/u-channel). The total amplitude is $\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$ (the minus sign comes from Fermi statistics — Griffiths Rule 8).

Squaring gives **four terms**:

$$\langle|\mathcal{M}|^2\rangle = \langle|\mathcal{M}_1|^2\rangle + \langle|\mathcal{M}_2|^2\rangle - \langle\mathcal{M}_1\mathcal{M}_2^*\rangle - \langle\mathcal{M}_1^*\mathcal{M}_2\rangle$$

Your tasks:

1. **Compute** the interference term $\langle\mathcal{M}_1\mathcal{M}_2^*\rangle$ by applying the Casimir-trick strategy (as given in the hint in Griffiths 7.37) to convert the spin sum into a single trace, then evaluate that trace using the contraction theorems.
2. **Do not re-derive** $\langle|\mathcal{M}_1|^2\rangle$ or $\langle|\mathcal{M}_2|^2\rangle$ — these can be read off from the electron–muon result (Griffiths Eq. 7.129 with $m = M = 0$). Simply state their values.
3. **Combine** all four terms into the full $\langle|\mathcal{M}|^2\rangle$, **rewrite** the result in terms of Mandelstam variables s, t, u.
4. **Obtain the cross section** $d\sigma/dt$ using the Lorentz-invariant formula.

§2 Background: What You Need Before Starting

§2.1 The Two Feynman Diagrams

Why Two Diagrams?

In electron–muon scattering ($e\mu \to e\mu$), there's only **one** diagram because the outgoing particles are distinguishable. But when both outgoing particles are electrons, we can't tell which outgoing electron came from which vertex. So we need two diagrams: one where $p_1 \to p_3$ (the "direct" or t-channel diagram, $\mathcal{M}_1$) and one where $p_1 \to p_4$ (the "exchange" or u-channel diagram, $\mathcal{M}_2$), which is $\mathcal{M}_1$ with $p_3 \leftrightarrow p_4$.

From Griffiths Eq. 7.107, the total amplitude is $\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$, where:

$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$$

$$\mathcal{M}_2 = -\frac{g_e^2}{(p_1 - p_4)^2}[\bar{u}(4)\gamma^\mu u(1)][\bar{u}(3)\gamma_\mu u(2)]$$

$\mathcal{M}_2$ is obtained from $\mathcal{M}_1$ by swapping $p_3 \leftrightarrow p_4$ (and $s_3 \leftrightarrow s_4$).

§2.2 Mandelstam Variables

Mandelstam Variables for $1 + 2 \to 3 + 4$

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \qquad \text{(total CM energy squared)} \tag{1}$$
$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 \qquad \text{(momentum transfer, direct)} \tag{2}$$
$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 \qquad \text{(momentum transfer, exchange)} \tag{3}$$

They satisfy: $s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2$.
For **massless** electrons: $\boxed{s + t + u = 0}$, so $u = -s - t$.
The propagator denominators are: $\mathcal{M}_1 \propto 1/t$, $\mathcal{M}_2 \propto 1/u$.

§2.3 Massless Dot Product Identities

Dot Products for $m = 0$ (from Griffiths' hint)

With all masses zero and four-momentum conservation $p_1 + p_2 = p_3 + p_4$:

$$s = 2(p_1 \cdot p_2) = 2(p_3 \cdot p_4) \tag{4}$$
$$t = -2(p_1 \cdot p_3) = -2(p_2 \cdot p_4) \tag{5}$$
$$u = -2(p_1 \cdot p_4) = -2(p_2 \cdot p_3) \tag{6}$$

These follow directly from expanding $(p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = 0 + 0 + 2p_1 \cdot p_2 = s$, etc.

§2.4 The Cross Section Formula

Lorentz-Invariant $d\sigma/dt$ Formula

For $2 \to 2$ scattering of massless particles:

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s\, p_i^{*2}} \langle |\mathcal{M}|^2 \rangle$$

where p_i^* is the CM momentum of the initial-state particles. For massless particles, $p_i^* = \sqrt{s}/2$, so $p_i^{*2} = s/4$, giving:

$$\boxed{\frac{d\sigma}{dt} = \frac{1}{16\pi s^2} \langle |\mathcal{M}|^2 \rangle}$$

We also need the statistical factor $S = 1/2$ for identical final-state particles (two electrons), which divides the final cross section by 2.

§3 Step 1: The "Easy" Terms — $\langle|\mathcal{M}_1|^2\rangle$ and $\langle|\mathcal{M}_2|^2\rangle$

What We're Doing

$\langle|\mathcal{M}_1|^2\rangle$ has the *same structure* as electron–muon scattering (two distinct currents connected by a single photon propagator). So we can read off the answer from Griffiths Eq. 7.129 with $m = M = 0$.

From Eq. 7.129 with $m = M = 0$:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{(p_1 \cdot p_3)^2}\left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\right]$$

Substituting the Mandelstam relations: $(p_1 \cdot p_3) = -t/2$, $(p_1 \cdot p_2) = s/2$, $(p_3 \cdot p_4) = s/2$, $(p_1 \cdot p_4) = -u/2$, $(p_2 \cdot p_3) = -u/2$:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{t^2/4}\left[\frac{s^2}{4} + \frac{u^2}{4}\right]$$

$\langle|\mathcal{M}_1|^2\rangle$

$$\boxed{\langle|\mathcal{M}_1|^2\rangle = 2g_e^4\,\frac{s^2+u^2}{t^2}}$$

For $\langle|\mathcal{M}_2|^2\rangle$, swap $t \leftrightarrow u$ (since $\mathcal{M}_2$ is $\mathcal{M}_1$ with $p_3 \leftrightarrow p_4$, which exchanges t and u):

$\langle|\mathcal{M}_2|^2\rangle$

$$\boxed{\langle|\mathcal{M}_2|^2\rangle = 2g_e^4\,\frac{s^2+t^2}{u^2}}$$

§4 Step 2: The Interference Term — $\langle\mathcal{M}_1\mathcal{M}_2^*\rangle$

This is the heart of the problem. The interference term has a different structure from the direct terms because spinors from *both* diagrams get tangled together.

§4.1 Writing $\mathcal{M}_1\mathcal{M}_2^*$

$$\mathcal{M}_1\mathcal{M}_2^* = \frac{g_e^4}{(p_1-p_3)^2(p_1-p_4)^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]\;[\bar{u}(4)\gamma^\nu u(1)]^*[\bar{u}(3)\gamma_\nu u(2)]^*$$

Using $[\bar{u}(a)\Gamma u(b)]^* = \bar{u}(b)\bar{\Gamma}u(a)$ with $\bar{\gamma}^\nu = \gamma^\nu$:

$$= \frac{g_e^4}{t\,u}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(1)\gamma^\nu u(4)]\;[\bar{u}(4)\gamma_\mu u(2)][\bar{u}(2)\gamma_\nu u(3)]$$

Notice the Spinor Chain

Unlike the direct terms (which factor into two independent traces), here the four spinor bilinears form a **single closed chain**: $3 \to 1 \to 4 \to 2 \to 3$. This means the spin sum will give **one** trace of eight gamma matrices, not a product of two traces of four.

§4.2 Applying the "Casimir-Like" Trick

Following Griffiths' hint, write out the indices explicitly and use the completeness relations $\sum_s u^{(s)}(p)\bar{u}^{(s)}(p) = \not{p}$ (since $m = 0$).

The sum over all four spins converts each $u\bar{u}$ pair into a $\not{p}$:

$$\sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^* = \frac{g_e^4}{t\,u}\ \mathrm{Tr}\Big[\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu\Big]$$

How to Read Off the Trace

Follow the spinor chain backwards around the loop $3 \to 1 \to 4 \to 2 \to 3$, writing down each $\not{p}$ and vertex factor as you encounter them, then close the loop with a trace. The ordering inside the trace is:

$$\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu$$

This is a trace of **8 gamma matrices** (each $\not{p} = \gamma^\alpha p_\alpha$ counts as one).

Including the spin-averaging factor of $1/4$:

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,t\,u}\ \mathrm{Tr}\Big[\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu\Big]$$

§4.3 Evaluating the Trace via Contraction Theorems

Strategy

Rather than expanding an 8-gamma trace directly (which would be brutal), we use the **contraction theorems** to simplify. Contract the μ index first (between γ^μ and γ_μ), then contract the ν index.

§4.3.1 Step 2a: Contract $\gamma^\mu \ldots \gamma_\mu$

Using the contraction theorem $\gamma_\mu\,\gamma^\alpha\gamma^\beta\gamma^\delta\,\gamma^\mu = -2\gamma^\delta\gamma^\beta\gamma^\alpha$ (Griffiths Rule 9), applied to the object between γ^μ and γ_μ:

$$\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu = -2\,\not{p}_4\,\gamma^\nu\,\not{p}_1$$

The trace becomes:

$$\mathrm{Tr}[\not{p}_3\,(-2\,\not{p}_4\,\gamma^\nu\,\not{p}_1)\,\not{p}_2\,\gamma_\nu] = -2\ \mathrm{Tr}[\not{p}_3\,\not{p}_4\,\gamma^\nu\,\not{p}_1\,\not{p}_2\,\gamma_\nu]$$

§4.3.2 Step 2b: Contract $\gamma^\nu \dots \gamma_\nu$

Now contract γ^ν with γ_ν. The object between them is $\not{p}_1 \not{p}_2$. Using $\gamma_\nu \gamma^\alpha \gamma^\beta \gamma^\nu = 4(a \cdot b)$ (Griffiths Rule 8′):

$$\gamma^\nu \not{p}_1 \not{p}_2 \gamma_\nu = 4(p_1 \cdot p_2)$$

So:

$$-2 \operatorname{Tr}[\not{p}_3 \not{p}_4 \cdot 4(p_1 \cdot p_2)] = -8(p_1 \cdot p_2) \operatorname{Tr}[\not{p}_3 \not{p}_4]$$

§4.3.3 Step 2c: Evaluate the Final Trace

Using $\operatorname{Tr}[\not{p}_3 \not{p}_4] = 4(p_3 \cdot p_4)$:

$$-8(p_1 \cdot p_2) \cdot 4(p_3 \cdot p_4) = -32(p_1 \cdot p_2)(p_3 \cdot p_4)$$

§4.4 Putting It Together

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,t\,u} \cdot (-32)(p_1 \cdot p_2)(p_3 \cdot p_4) = \frac{-8g_e^4 (p_1 \cdot p_2)(p_3 \cdot p_4)}{t\,u}$$

Substituting $(p_1 \cdot p_2) = s/2$ and $(p_3 \cdot p_4) = s/2$:

Interference Term

$$\boxed{\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{-2g_e^4\, s^2}{t\,u}}$$

Since $\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle^*$, and this result is **real**, we have $\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$.

§5 Step 3: The Full Spin-Averaged Amplitude Squared

$$\langle |\mathcal{M}|^2 \rangle = \langle |\mathcal{M}_1|^2 \rangle + \langle |\mathcal{M}_2|^2 \rangle - \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle - \langle \mathcal{M}_1^* \mathcal{M}_2 \rangle$$

$$= 2g_e^4 \frac{s^2 + u^2}{t^2} + 2g_e^4 \frac{s^2 + t^2}{u^2} - 2 \times \left(\frac{-2g_e^4 s^2}{tu} \right)$$

Full Spin-Averaged $|\mathcal{M}|^2$ for Møller Scattering ($m = 0$)

$$\boxed{\langle |\mathcal{M}|^2 \rangle = 2g_e^4 \left[\frac{s^2 + u^2}{t^2} + \frac{s^2 + t^2}{u^2} + \frac{4s^2}{tu} \right]}$$

Structure of the Result

The first two terms come from the squared diagrams individually; each looks like the electron–muon result with a t or u channel propagator. The third term is the **interference** between the two diagrams. Note that the interference is **positive** (the two minus signs from $-\mathcal{M}_2$ and from the $-2g_e^4 s^2/tu$ result combine to give $+4s^2/tu$). This makes physical sense: the two diagrams interfere constructively.

§6 Step 4: The Differential Cross Section

Using the formula $d\sigma/dt = \langle|\mathcal{M}|^2\rangle/(16\pi s^2)$, and including the identical-particle factor $S = 1/2$:

$$\frac{d\sigma}{dt} = \frac{S}{16\pi s^2}\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\cdot\frac{2g_e^4}{16\pi s^2}\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]$$

Now substitute $g_e^2 = 4\pi\alpha$ (so $g_e^4 = 16\pi^2\alpha^2$):

$$\frac{d\sigma}{dt} = \frac{16\pi^2\alpha^2}{16\pi s^2}\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]$$

Differential Cross Section for Møller Scattering ($m = 0$)

$$\boxed{\frac{d\sigma}{dt} = \frac{\pi\alpha^2}{s^2}\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu}\right]}$$

with $u = -s - t$. This gives $d\sigma/dt$ as a function of s and t alone.

§7 Summary: The Calculation Pipeline

Step	Tool Used	Result
$\langle\|\mathcal{M}_1\|^2\rangle$	Read from Eq. 7.129 (m=0)	$2g_e^4(s^2+u^2)/t^2$
$\langle\|\mathcal{M}_2\|^2\rangle$	Swap $t \leftrightarrow u$ in above	$2g_e^4(s^2+t^2)/u^2$
$\langle\mathcal{M}_1\mathcal{M}_2^*\rangle$	Single trace of 8 γ's, contracted via Rules 8′ & 9′	$-2g_e^4 s^2/(tu)$
Combine with signs	$\|\mathcal{M}_1 - \mathcal{M}_2\|^2$	Sum of 3 terms
Cross section	$d\sigma/dt$ formula + $S = 1/2$	$(\pi\alpha^2/s^2)[\ldots]$

The Big Takeaway

The interference term is the only genuinely new calculation. Unlike the direct/exchange terms (which factored into products of two 4-gamma traces), the interference produces a **single** 8-gamma trace. The contraction theorems let you collapse this 8-gamma trace in two steps rather than expanding it with the 4-gamma trace theorem, which would be far more tedious.

The physical lesson: identical particles in the final state generate interference terms that are computationally harder than the individual squared diagrams. This is the price you pay for Fermi statistics.

Problem H4, Griffiths 7.37 (parts).
Sources: Griffiths Ch. 7, Thomson Ch. 6, Lancaster & Blundell Ch. 40.